

VIVAAN THADANI

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SUMMARY OF QUALIFICATIONS

- Mechanical engineering student with hands-on experience in CAD modelling, mechanical drawings, rapid prototyping, design validation, and structural analysis.
- Created and modified 3D CAD models using SolidWorks and AutoCAD to support prototype development, assembly planning, and fabrication workflows.
- Supported R&D-style engineering projects through physical testing, wind tunnel testing, iterative design, technical documentation, and performance improvement.
- Organized engineering reports, technical data, and preliminary BOM-style documentation to improve design communication across team-based projects.

SKILLS

- **Engineering Software:** SolidWorks, ANSYS, Fusion 360, AutoCAD, Bambu Lab
- **Core Engineering Skills:** CAD Modelling, Mechanical Drawings, GD&T, Design Validation, Structural Analysis
- **Programming and Data Tools:** MATLAB, Python, C, Microsoft Excel

EDUCATION

Toronto Metropolitan University

Sep 2024 – May 2029 (Expected)

Bachelor of Engineering, Mechanical Engineering, cGPA: 3.83/4.33

Dean's List: 2024, 2025

- **Relevant Coursework:** Manufacturing, Engineering Design, Statics, Mechanics of Materials, Dynamics, Thermodynamics, Circuits

EXPERIENCE + PROJECTS

Mechanical Engineering Intern – Toronto Water

May 2026 – Aug 2026

- Design and created a solar powered wiper for a live feed sewer camera using Fusion 360, Arduino and 3D printing for the sewer environment that resulted in a maintenance cost reduction of \$6000 annually per camera site
- Design and fabricate custom sensor caps using CAD software, 3D printing, and CNC machining to ensure secure and reliable installation of monitoring equipment in confined storm water and sewage tunnel environments.
- Proposed and created user guides and workflow documentation for the inventory system to develop systems for maintaining as well as tracking sensors across the city and trained over 40 operators accordingly.

Structures Team Member – TMU Hyperloop

Oct 2025 – Present

- Designed and fabricated a 3D-printed NACA duct using SolidWorks, rapid prototyping, and wind tunnel testing to support airflow-based design validation.
- Prepared CAD models, mechanical drawings, and technical documentation to improve prototype fabrication, assembly readiness, and design communication.
- Coordinated mechanical and electrical interfaces across vehicle systems to improve subsystem compatibility and support cross-functional integration.

Summer Camp Counsellor – Town of Bradford

Summer 2023 – Summer 2025

- Supervised 20–30 campers during structured activities, maintaining a safe, inclusive, and engaging environment for children with diverse needs.
- Designed and led games, crafts, and group activities while communicating behavioural concerns to parents and camp leadership as needed.